01 ប្រមាណវិធីលើរ៉ាឌីកាល់សន្ទស្សន៍ 2 និង 3

1. បើ $a>0, x^2=a$ នោះ $x=\pm\sqrt{a}$ ។
2. បើ x^3 នោះ $x=\sqrt[3]{a}$ដែល a អាចធំឬតូចជាងសូន្យ។
3. លក្ខណៈ: បើ a,b ជាចំនួនពិត និង $b\neq 0$

$\sqrt{a}\cdot\sqrt{b}=\sqrt{ab}$ $\qquad$ $\sqrt[3]{a}\cdot\sqrt[3]{b}=\sqrt[3]{ab}$

$\dfrac{\sqrt{a}}{\sqrt{b}}=\sqrt{\dfrac{a}{b}}$ $\qquad$ $\dfrac{\sqrt[3]{a}}{\sqrt[3]{b}}=\sqrt[3]{\dfrac{a}{b}}$

4. ដើម្បីបូកឬដកកន្សោមមានរ៉ាឌីកាល់គេត្រូវបំលែងអោយសន្ទស្សន៍និងរ៉ាឌីកង់អោយដូចគ្នា រួច បូកឬដក តែមេគុណ។
5. បំបាត់រ៉ាឌីកាល់ភាគបែង :

រំលឹក៖ $(a-b)(a+b)=a^2-b^2$

$a^3\pm b^3=(a\pm b)(a^3\mp a^b+ab^2\mp b^3)$

ករណីភាគបែងមានពីរតួ ជាឬសការេ ត្រូវគុណនិងកន្សោមផ្លាស់។

$$\frac{1}{\sqrt{7}-1}=\frac{\sqrt{7}+1}{(\sqrt{7}-1)(\sqrt{7}+1)}=\frac{\sqrt{7}+1}{7-1}=\frac{\sqrt{7}+1}{6}$$

ករណីភាគបែងមានពីរតួ ជាឬសគូប ត្រូវគុណកន្សោមបំពេញ។

$$\frac{1}{\sqrt[3]{2}+1}=\frac{1\cdot(\sqrt[3]{4}-\sqrt[3]{2}+1)}{(\sqrt[3]{2}+1)(\sqrt[3]{4}-\sqrt[3]{2}+1)}=\frac{\sqrt[3]{4}-\sqrt[3]{2}+1}{3}$$

លំហាត់

01 គណនា

ក. $\sqrt{9}$ $\quad$ ខ. $\sqrt{16}$ $\quad$ គ. $\sqrt{36}$ $\quad$ ឃ. $-\sqrt{64}$ $\quad$ ង. $-\sqrt{100}$

ច. $\sqrt{121}$ $\quad$ ឆ. $-\sqrt{144}$ $\quad$ ជ. $\sqrt{625}$ $\quad$ ឈ. $\sqrt[3]{8}$ $\quad$ ញ. $\sqrt[3]{-8}$

ដ. $-\sqrt[3]{27}$ $\quad$ ឋ. $-\sqrt[3]{64}$ $\quad$ ឌ. $\sqrt[3]{125}$ $\quad$ ឍ. $\sqrt[3]{216}$ $\quad$ ណ. $\sqrt[3]{100}$

ត. $\sqrt{\dfrac{9}{16}}$ $\quad$ ថ. $\sqrt{\dfrac{49}{9}}$ $\quad$ ទ. $-\sqrt{\dfrac{81}{4}}$ $\quad$ ធ. $\sqrt{\dfrac{169}{49}}$ $\quad$ ន. $\sqrt{\dfrac{196}{25}}$

ប. $-\sqrt{\dfrac{400}{225}}$ $\quad$ ផ. $\sqrt[3]{\dfrac{1}{8}}$ $\quad$ ព. $\sqrt[3]{\dfrac{8}{27}}$ $\quad$ ភ. $3\sqrt{\dfrac{64}{125}}$ $\quad$ ម. $\sqrt[3]{\dfrac{512}{343}}$

02 សម្រួលរ៉ាឌីកាល់

ក. $\sqrt{16^3}$ $\quad$ ខ. $-\sqrt[3]{36^2}$ $\quad$ គ. $\sqrt{64^3}$ $\quad$ ឃ. $(\sqrt[3]{-8})^3$ $\quad$ ង. $\sqrt[3]{-27^3}$

ច. $\sqrt[3]{15}$ $\quad$ ឆ. $\sqrt[3]{8^2}$ $\quad$ ជ. $\sqrt[3]{64^2}$ $\quad$ ឈ. $\sqrt[3]{(-27)^2}$ $\quad$ ញ. $\sqrt{y^2}$

ដ. $\sqrt{x^4}$ $\quad$ ឋ. $\sqrt{x^2y^4}$ $\quad$ ឌ. $\sqrt{y^6}$ $\quad$ ឍ. $\sqrt{\dfrac{16}{x^2}}$ $\quad$ ណ. $\sqrt{\dfrac{100}{n^4}}$

03 សម្រួលកន្សោម

ក. $\sqrt{(2x)^2}$ $\quad$ ខ. $\sqrt[3]{(-5y)^3}$

គ. $\sqrt{(4-a)^2}$ $\quad$ ឃ. $\sqrt[3]{(x+3)^3}$

ង. $\sqrt{16b^2+24b+9}$ $\quad$ ច. $\sqrt{9x^2-30x+25}$

ឆ. $\sqrt{4m^2-20mn+25n^2}$ $\quad$ ជ. $\sqrt{49x^2-112xy+64y^2}$

04 បញ្ចេញមួយចំនួនពីរ៉ាឌីកាល់

ក. $\sqrt{18}$ $\quad$ ខ. $-\sqrt{48}$ $\quad$ គ. $\sqrt{75}$ $\quad$ ឃ. $\sqrt{\dfrac{30}{49}}$

ង. $\sqrt{\dfrac{10}{121}}$ $\quad$ ច. $\sqrt[3]{40}$ $\quad$ ឆ. $\sqrt[3]{54}$ $\quad$ ជ. $-\sqrt[3]{128}$

ឈ. $\sqrt[3]{192}$ $\quad$ ញ. $\sqrt[3]{\dfrac{3m}{8n^3}}$ $\quad$ ដ. $\sqrt[3]{16a^5}$ $\quad$ ឋ. $\sqrt{36a^2b^3}$

ឌ. $\sqrt{27a^4b^3}$ $\quad$ ឍ. $\sqrt{72x^5y^2}$ $\quad$ ណ. $\sqrt{80m^4n^3}$ $\quad$ ត. $-\sqrt{112a^3b^4}$

ថ. $\sqrt{64x^2y^3}$ $\quad$ ទ. $\sqrt[3]{16m^3n^3}$ $\quad$ ធ. $\sqrt[3]{-54x^4b^3}$ $\quad$ ន. $-\sqrt[3]{128a^5y^3}$

05 បញ្ចូលមួយចំនួនក្នុងរ៉ាឌីកាល់ x,y,m,nនិង pជាចំនួនវិជ្ជមាន

ក. $5\sqrt{6}$ $\quad$ ខ. $2m\sqrt{m}$ $\quad$ គ. $\dfrac{\sqrt{23}}{y^3}$ $\quad$ ឃ. $2\sqrt[3]{5}$

ង. $2x\sqrt[3]{4}$ $\quad$ ច. $\sqrt[3]{\dfrac{3m}{2n}}$

06 គណនា

ក. $3\sqrt{2}-4\sqrt{2}+5\sqrt{2}-3\sqrt{2}$ $\quad$ ខ. $5\sqrt{2}-3\sqrt{3}-6\sqrt{2}+5\sqrt{3}$

គ. $3\sqrt{15}-4\sqrt{3}-3\sqrt{15}+6\sqrt{3}$ $\quad$ ឃ. $4\sqrt{3}-2\sqrt{17}+3\sqrt{17}-3\sqrt{3}-2\sqrt{3}$

ង. $2\sqrt[3]{2}-8\sqrt[3]{3}+\sqrt[3]{2}+3\sqrt[3]{3}$ $\quad$ ច. $8\sqrt[3]{2}-3\sqrt[3]{3}-5\sqrt[3]{2}+2\sqrt[3]{3}$

07 គណនា

ក. $\dfrac{2}{3}\sqrt{27}-\dfrac{3}{4}\sqrt{48}$ $\quad$ ខ. $\dfrac{1}{4}\sqrt{288}-\dfrac{1}{6}\sqrt{72}$

គ. $\dfrac{3}{5}\sqrt{75}-\dfrac{2}{3}\sqrt{27}$ $\quad$ ឃ. $5\sqrt[3]{128}-3\sqrt[3]{250}$

ង. $3\sqrt[3]{81}-\dfrac{1}{2}\sqrt[3]{192}$ $\quad$ ច. $4\sqrt[3]{54}-3\sqrt[3]{128}$

08 គណនា

ក. $2\sqrt{8}-3\sqrt{98}-2\sqrt{200}$ $\quad$ ខ. $-3\sqrt{50}-\sqrt{32}+5\sqrt{200}$

គ. $3\sqrt{175}-2\sqrt{28}+3\sqrt{63}-\sqrt{112}$ $\quad$ ឃ. $\sqrt{108}-2\sqrt{27}-\sqrt{40}-5\sqrt{160}$

ង. $2\sqrt[3]{16}+3\sqrt[3]{54}-2\sqrt[3]{128}$ $\quad$ ច. $3\sqrt[3]{81}+\dfrac{1}{2}\sqrt[3]{128}-3\sqrt[3]{192}+4\sqrt[3]{54}$

ឆ. $4\sqrt[3]{54}-6\sqrt[3]{81}-4\sqrt[3]{16}+3\sqrt[3]{24}$ $\quad$ ជ. $-2\sqrt[3]{40}-3\sqrt[3]{135}+5\sqrt[3]{320}+8\sqrt[3]{5}$

09 គណនា

ក. $-2(2\sqrt{12}-\sqrt{18})-5(3\sqrt{32}-\sqrt{27})$ $\quad$ ខ. $3(3\sqrt[3]{40}-\sqrt[3]{135})+4(\sqrt[3]{320}-\sqrt[3]{40})$

គ. $\dfrac{2\sqrt{27}}{3}-3\sqrt{48}+\dfrac{4\sqrt{50}}{5}-\dfrac{4\sqrt{18}}{3}$ $\quad$ ឃ. $\dfrac{2}{3}\sqrt[3]{81}-\dfrac{1}{2}\sqrt[3]{24}+\dfrac{2\sqrt[3]{135}}{3}-\dfrac{3\sqrt[3]{40}}{2}$

10 គណនាកន្សោមខាងក្រោម a,b,x,y និង z ជាចំនួនវិជ្ជមាន

$A=-3\sqrt{32x}+6\sqrt{8x}$ $\quad$ $B=2\sqrt{125x^2z}+8x\sqrt{80z}$

$C=7a\sqrt{b^3}+b\sqrt{4a^2b}-\sqrt{4b}$ $\quad$ $D=8b\sqrt{49b}-7\sqrt{9b^3}+a\sqrt{4a}+\sqrt{a^3}$

$E=3xy\sqrt{x^2y}-2\sqrt{x^4y^3}$ $\quad$ $F=-3a\sqrt{a^3b^5}-2b\sqrt{a^5b^3}+5\sqrt{a^5b^3}$

$G=8a\sqrt[3]{54a}+6\sqrt[3]{16a^4}$ $\quad$ $H=3\sqrt[3]{x^4y}-6x\sqrt[3]{xy^4}+2\sqrt[3]{x^4y^4}$

11 គណនាតម្លៃលេខនៃកន្សោម

❶ $A=\sqrt{ab}-\sqrt{ab^3}-\sqrt{9a^3b^3}-\sqrt{a^3b}$ ចំពោះ $a=5, b=3$

❷ $A=\sqrt{4a}+a\sqrt{a^2b}+\sqrt{b^2a}+b\sqrt{9b}$ ចំពោះ $a=3, b=2$

12 គណនា

ក.$(2\sqrt{3})(3\sqrt{2})$ ខ.$(4\sqrt{6})(-2\sqrt{5})$ គ.$(3\sqrt{5})(5\sqrt{3})$

ឃ.$(6\sqrt{2})(-12\sqrt{3})$ ង.$(3\sqrt[3]{8})(-3\sqrt{48})$ ច.$(-3\sqrt{75})(-2\sqrt{48})$

ឆ.$(2\sqrt[3]{3})\left(-\frac{1}{2}\sqrt[3]{2}\right)$ ជ.$(3\sqrt[3]{2})(5\sqrt[3]{15})$ ឈ.$(6\sqrt[3]{8})(-3\sqrt[3]{2})$

ញ.$3\sqrt{5}(2\sqrt{18}-3\sqrt{48})$ ដ.$-3\sqrt{3}(3\sqrt{6}-3\sqrt{2})$ ឋ.$\frac{1}{2}\sqrt{3}(2\sqrt{48}-3\sqrt{32})$

ឌ.$\frac{3}{2}\sqrt{2}(2\sqrt{18}-3\sqrt{48})$ ឍ.$-4\sqrt[3]{3}(2\sqrt[3]{6}-2\sqrt[3]{5})$ ណ.$2\sqrt[3]{5}(3\sqrt[3]{3}-5\sqrt[3]{2})$

13 គណនា

ក.$(2\sqrt{3}-8)(9+2\sqrt{5})$ ខ.$(3\sqrt{5}-2\sqrt{10})(\sqrt{50}-2\sqrt{80})$

គ.$(\sqrt{50}-\sqrt{75})(\sqrt{32}-\sqrt{48})$ ឃ.$(\sqrt{125}-\sqrt{75})(\sqrt{80}-\sqrt{48})$

ង.$(3\sqrt[3]{18}+3\sqrt[3]{27})(2\sqrt[3]{8}-2\sqrt[3]{12})$ ច.$(3\sqrt[3]{80}-2\sqrt[3]{27})(-3\sqrt[3]{20}-3\sqrt[3]{12})$

14 គេឱ្យ $a=3\sqrt{5}-2\sqrt{10}$,$b=5\sqrt{7}+2\sqrt{10}$, $c=\sqrt[3]{18}-\sqrt[3]{27}$ និង $d=3\sqrt[3]{6}+\sqrt[3]{8}$ ។ គណនា

❶ $-3ab$ ❷ a^2+b^2 ❸ a^2-2b^2

❹ b^2-2ab ❺ $\frac{1}{2}cd$ ❻ c^2-b^2

15 គណនា

ក. $\frac{\sqrt{33}}{11}$ ខ. $\frac{-\sqrt{48}}{\sqrt{27}}$ គ. $2\sqrt{75}\div\sqrt{15}$

ឃ. $\frac{2\sqrt{84}}{-\sqrt{12}}$ ង. $\frac{5\sqrt{25}}{\sqrt{75}}\div\sqrt{5}$ ច. $\frac{12\sqrt[3]{6}}{\sqrt[3]{2}}$

ឆ. $\frac{5\sqrt[3]{100}}{25\sqrt[3]{5}}$ ជ. $\left(\frac{\sqrt[3]{72}}{2\sqrt[3]{8}}\right)\times\sqrt[3]{2}$ ឈ. $\frac{6\sqrt[3]{7}}{3\sqrt[3]{6}}\div\frac{4\sqrt[3]{21}}{2\sqrt[3]{3}}$

16 សម្រួលកន្សោម

$A=\sqrt{\frac{b^2}{b^2-14b+49}}$ $B=\sqrt{\frac{49x^2-56x+16}{36x^2}}$

$C=\sqrt{\frac{a^2+16ab+64b^2}{a^2+10ab+25b^2}}$ $D=\sqrt{\frac{25b^2+10ab+a^2}{16b^2+24ab+9a^2}}$

17 បំបាត់ភាគបែងនៃកន្សោមខាងក្រោម

ក. $\frac{3}{\sqrt{10}}$ ខ.$\frac{3\sqrt{3}}{\sqrt{33}}$ គ.$\frac{6}{\sqrt{48}}$ ឃ.$\frac{8}{\sqrt{27}}$ ង.$\frac{9\sqrt{4}}{\sqrt{18}}$

ច.$\frac{5\sqrt{5}}{\sqrt{72}}$ ឆ.$\frac{\sqrt[3]{9}}{\sqrt[3]{10}}$ ជ.$\frac{2\sqrt[3]{3}}{\sqrt[3]{30}}$ ឈ.$\frac{\sqrt[3]{18}}{\sqrt[3]{20}}$ ញ.$\sqrt[3]{\frac{7m}{36n}}$

ដ.$\sqrt[3]{\frac{11p}{49q}}$ ឋ.$\sqrt[3]{\frac{3}{4y^2}}$ ឌ.$\frac{36-\sqrt{6}}{\sqrt{8}}$ ឍ.$\frac{\sqrt{3}+\sqrt{5}}{3\sqrt{20}}$ ណ.$\frac{8}{2\sqrt{75}-3\sqrt{50}}$

ត.$\frac{2\sqrt{3}}{2\sqrt{80}-\sqrt{45}}$ ថ.$\frac{9-\sqrt[3]{3}}{2\sqrt[3]{32}}$ ទ.$\frac{5\sqrt[3]{4}+\sqrt[3]{3}}{8\sqrt[3]{13}}$

18 បំបាត់ភាគបែងដែលអថេរជាចំនួនវិជ្ជមាន

ក. $\frac{3}{\sqrt{x}}$ ខ.$\frac{3a}{\sqrt{8ab}}$ គ. $\frac{3\sqrt{x}}{5\sqrt{2x^3y}}$ ឃ.$\frac{8a}{\sqrt[3]{25ab}}$

ង.$\frac{27\sqrt[3]{m}}{7\sqrt[3]{m^2n}}$ ច.$\frac{15}{\sqrt{2a}-\sqrt{b}}$ ឆ.$\frac{7\sqrt{2x}}{2\sqrt{x}+\sqrt{3y}}$ ជ.$\frac{\sqrt[3]{11}}{\sqrt[3]{2m}-\sqrt[3]{12n}}$

20 គេឱ្យត្រីកោណកែងមួយមានអៃក្រ a, b, c ដែល c ជាប្រវែងអ៊ីប៉ូទេនុស ។គណនាជ្រុងមួយនៃត្រីកោណកែងនេះ បើគេដឹងថា

❶ $a=3, b=4$ ❷ $a=5, b=12$

❸ $a=6, b=8$ ❹ $c=26, b=10$

❺ $b=30, c=50$

21 គេឱ្យ $m=3\sqrt{8}+\sqrt{5}$ និង $n=3\sqrt{8}-\sqrt{5}$ ។ គណនា

❶ $\frac{mn+m^2}{m}$ ❷ $\frac{m^2-n^2}{m+n}$ ❸ $\frac{n^2-2mn}{n^2}$

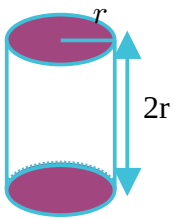
22 ប្រៀបធៀបចំនួន $\sqrt{2}$ និង $\sqrt[3]{3}$

23 សម្រួលកន្សោម

$A=\sqrt{22-\sqrt{288}}$ $B=\sqrt{7-2\sqrt{12}}$ $C=\sqrt{5+\sqrt{24}}$

$D=\sqrt{7+\sqrt{13}}$ $E=\sqrt{3+\sqrt{5}}$ $F=\sqrt{6+\sqrt{11}}$

24 គណនាកាំ r នៃស៊ីឡាំង បើឌីណាម៉ែត្រមានរង្វាស់ $628\,000\ \text{cm}^3$ គេឱ្យ $\pi=3.14$



25 ❶ បង្ហាញថា $2^{17}+2^{17}=2^{18}$

❷ កំណត់ x ដែល $2^x\cdot 2^{x+3}=8^5$